Amsterdam University College
Text Mining
Jelke Bloem

Project Report

From Totoro to Tangled: analyzing lexical and thematic patterns in Studio Ghibli vs Disney

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Abstract:
This study applies computational text analysis to investigate linguistic and emotional differences in dialogue from animated films produced by Disney and Studio Ghibli. Using natural language processing techniques, including topic modeling (LDA), transformer-based sentiment analysis, and zero-shot thematic classification, we analyze film subtitles to uncover recurring themes, emotional arcs, and gendered language patterns. Our findings suggest that Studio Ghibli emphasizes themes such as grief and friendship, with more balanced gender representation, while Disney highlights loneliness and identity, often featuring male-coded language and conflict-driven topics. These results reveal measurable narrative distinctions between the studios and demonstrate how NLP can contribute to film analysis by uncovering subtle but meaningful storytelling patterns.

1. Introduction:

Disney and Studio Ghibli are two of the most influential studios in the history of animated cinema. Both have produced films that are widely celebrated around the world, yet their approaches to storytelling differ in notable ways. Disney is often associated with structured narratives that revolve around heroic protagonists, romance, and musical sequences. In contrast, Studio Ghibli is known for its slower narrative pace, emotional subtlety, and recurring themes related to nature, everyday life, and the spiritual world.

The aim of this project is to investigate how these stylistic and thematic differences are reflected in the language used within each studio's film dialogue. Specifically, we ask: How do the most common emotions and prevailing topics in Ghibli dialogue compare to those in Disney dialogue? To explore this question, we apply a combination of natural language processing methods to the subtitle data of selected films from both studios. These include Latent Dirichlet Allocation (LDA) for topic modeling, transformer-based sentiment analysis, and zero-shot classification to identify deeper thematic patterns. We also analyze the distribution of gendered language across topics as a proxy for character representation. Through this comparative analysis, we aim to identify measurable linguistic and emotional patterns that may distinguish the narrative styles of these two iconic studios.

2. Related Work:

The analysis of movie scripts and dialogue as unstructured text has gained significant attention in recent years, with researchers exploring various computational approaches to understand narrative patterns, emotional arcs, and thematic elements in cinematic storytelling.

2.1 Topic Modelling

The application of topic modeling to narrative content has proven effective for identifying thematic patterns across large corpora. Chao and Sirmorya (2016) demonstrated the viability of Latent Dirichlet Allocation (LDA) for automated movie genre classification, achieving F-scores of up to 0.49 by modeling thematic patterns in scripts. Their work showed that LDA-based topic modeling could compete with visual and audio-based classification methods while being computationally less expensive.

The integration of topic modeling with recommendation systems has also shown promise. Fasano's movie recommendation system (2022) combined collaborative filtering with LDA-derived topic vectors, creating a hybrid Collaborative Topic Model (CTM) that improved recommendation relevance and addressed cold-start problems for new movies. This work demonstrated that thematic information extracted through topic modeling adds meaningful signal to collaborative filtering approaches

. However, as noted in critical analyses of LDA applications (Diggit Magazine, 2023), topic modeling has inherent limitations when applied to discourse analysis. While LDA excels at identifying co-occurring word patterns in large corpora, it cannot capture semantic nuances, irony, or implicit meanings that are crucial in narrative analysis. The interpretation of generated topics always requires human intervention to ensure semantic coherence and meaningfulness.

Our implementation follows established best practices for LDA topic modeling while accounting for these limitations. We apply LDA separately to Disney and Studio Ghibli corpora to preserve each studio's unique linguistic characteristics, and we use coherence scores (C_v and NPMI measures) to determine optimal topic numbers.

2.2 Sentiment Analysis

Early computational script analysis established foundational approaches for narrative pattern recognition. Lee, Yu, and Cheong (2017) systematically analyzed 978 movie scripts using SentiWordNet, dividing scripts into 20-line blocks to identify six distinct narrative structures, though lexicon-based methods may miss contextual nuances. Gorinski (2017) developed ScriptBase with nearly 1,000 Hollywood scripts and LSTM-based summarization models combining textual and visual information.

Recent transformer-based approaches have addressed earlier limitations. Exner and Klinger (2021) applied deep learning sentiment analysis to movie subtitles for predicting emotional arcs, validating segmentation approaches despite challenges with humor and irony. Deldjoo et al. (2016) used MaxEnt classifiers on subtitle segments to detect emotional scenes, while Habernal and Konopík (2010) analyzed sentiment progression in scripts, noting positive bias in standard models. Ramakrishna et al. (2020) introduced MultiSentimentArcs for measuring sentiment coherence across film narratives, emphasizing consistency challenges in dialogue analysis.

Our approach builds upon these methodological foundations while addressing their limitations through the use of CardiffNLP's twitter-roberta-base-sentiment model. This RoBERTa-based model, fine-tuned on Twitter data, is particularly well-suited to the short, context-light nature of subtitle dialogue, representing a significant advancement over both lexicon-based methods and earlier neural approaches that were not optimized for brief, informal text segments. The model's effectiveness on short, context-limited text has been validated in benchmark evaluations such as TweetEval (Barbieri et al., 2020), making it an appropriate choice for subtitle-based sentiment analysis.

3. Topic Modelling

3.1 Preprocessing

The preprocessing pipeline for LDA included splitting each movie into six equal segments to increase the number of documents, aiming to improve model performance. To reduce noise, standard English stopwords, studio-specific character names, and common movie-related terms that frequently appeared in subtitles but lacked thematic value were removed.

Tokens were lemmatized using part-of-speech tagging and only nouns longer than two characters were retained. We then filtered the vocabulary by removing words that appeared in fewer than two documents or in more than 80% of documents. This ensured that the final input to the LDA model was both linguistically meaningful and consistent across the corpus. However, due to occasional inaccuracies in POS tagging, some non-nouns may still be present in the processed data.

3.2 LDA

LDA was used to generate topic models for both Studio Ghibli and Disney movie subtitles. Model quality was evaluated using coherence scores, which measure how semantically related the top words in each topic are; higher scores indicate more interpretable and coherent topics. Topic clusters were labeled by interpreting the most prominent, high-ranking words associated with each topic. To explore potential gender differences, we conducted a closer analysis by calculating the probabilities of gendered words within the topic distributions for each studio

4. Sentiment Analysis

To explore how emotional tone is expressed in Disney and Studio Ghibli films, we applied sentence-level sentiment analysis to their subtitle data. The goal was to investigate whether meaningful differences exist between the two studios in terms of how often positive, neutral, or negative sentiments are conveyed, and how those sentiments evolve over the course of a film's narrative. Because subtitle lines are typically short, informal, and often lack surrounding context, we selected a model suited to this structure: CardiffNLP's twitter-roberta-base-sentiment. While not specifically trained on scripts, this RoBERTa-based model was the most appropriate among existing options, having been fine-tuned on tweets (short, context-light text comparable to subtitle dialogue). We began by applying the model to every line of dialogue across all films and compared the resulting sentiment distributions between the two studios. For this second analysis, we divided each film's subtitles into 100 equal chronological segments and computed a net sentiment score per segment, defined as the difference between the proportions of positive and negative lines. These scores allowed us to visualize emotional fluctuation across each film's runtime.

5. Enhanced Analysis with Multi-Model Approach

Given the limitations of the initial sentiment analysis, most notably, the overwhelming dominance of neutral classifications, we expanded our methodology to better capture the emotional and thematic richness of animated film dialogue. A single polarity-based model proved inadequate for identifying the nuanced expressions present in short, context-light subtitle lines.

To address this, we implemented a dual-model pipeline. First, we replaced the original sentiment classifier with tabularisai/multilingual-sentiment-analysis, a model offering five sentiment categories: very negative, negative, neutral, positive, and very positive. This allowed for greater granularity while maintaining robustness on short textual inputs.

Second, we introduced zero-shot thematic classification using facebook/bart-large-mnli, which evaluates dialogue lines against a set of ten predefined emotional and thematic categories tailored to animated storytelling: Resilience, Grief, Trauma, Personal Growth, Empathy, Romantic Love, Friendship, Family, Loneliness, and Identity. This approach enables us to identify emotionally significant content that traditional sentiment analysis might miss.

5.1 Implementation Details

Our enhanced pipeline processes each subtitle line through both models in parallel. Sentiment predictions are handled with additional safeguards: if the model's confidence is below a 0.45 threshold, the line is labeled as "uncertain" rather than forcing a prediction. In cases where a line is syntactically a command or exclamation—identified through regular expressions—and receives a neutral classification, we flag it as "likely emotive." Meanwhile, the zero-shot classifier returns multiple candidate thematic labels, from which we select the highest-confidence match for primary analysis. This design allows us to retain interpretability while increasing the accuracy and depth of emotional and thematic insights derived from subtitle data.

6. Results and Findings

6.1 Topic Modelling

A comparison of coherence scores can be found in Appendix A. For Disney, the best coherence score was 0.3658 with 7 topics, while for Studio Ghibli the highest score was 0.4070 at 10 topics. Despite Ghibli's best performance at 10 topics, we opted to use 7 topics for both studios to ensure a more balanced and meaningful comparison. The coherence score for Ghibli at 7 topics was still relatively strong at 0.3911. The discovered topics and their top 8 words are displayed in Appendix B and C . While the Ghibli topics appeared thematically coherent and easier to interpret, the Disney topics were generally more ambiguous and harder to label clearly.

The distribution of topics across documents, shown in Appendix D, appears relatively balanced. However, it's important to note that each movie was split into six segments, which may give the illusion of uniformity and may influence topic frequency patterns. While some semantic themes stood out, such as Disney topics including more terms related to danger or threat (e.g., savage, predator) and Ghibli topics, particularly Topic 6, featuring nature-related elements consistent with the studio's storytelling style, we could not identify a strong or systematic thematic difference between the two studios based on topic modeling alone. Thematic distinctions were present but subtle and not consistent enough to suggest a clear division in narrative focus.

In terms of gender representation, we observed that male-gendered terms (e.g., prince, hero) were more prominent in Disney topics. In contrast, Ghibli topics featured more female-gendered terms overall. To quantify this observation, we conducted a further analysis comparing the probabilities of gendered words within the topic distributions of each studio. Words like man, prince, and dad appeared in both corpora, but with slightly higher probabilities in Disney. For example, prince had a probability of 0.0085 in Disney Topic 1, while in Ghibli it peaked at 0.0038 in Topic 4. Ghibli also showed higher values for female terms such as mom, princess, and woman, with princess reaching 0.0290 in Ghibli Topic 6. The full breakdown of these gender word probabilities can be found in Appendix E.

6.2 Sentiment Analysis

As can be seen in Appendix F, the sentiment distribution across all subtitle lines is heavily skewed toward neutral sentiment for both Disney and Ghibli films. Positive and negative sentiments appear in much smaller proportions, with minimal variation between the two studios. This suggests that subtitle dialogue tends to be emotionally neutral or lacks sufficient context for strong sentiment classification. To examine emotional arcs, we divided each film's subtitles into 100 equal chronological segments and calculated net sentiment scores (positive minus negative). As shown in Appendix G, the results revealed no consistent or interpretable emotional trends across films or between studios. Emotional fluctuations appeared irregular and varied widely between movies, likely due to the dominance of neutral classifications and the fragmented nature of subtitle dialogue.

6.3 Zero-Shot Thematic Classification Results

The zero-shot classification revealed distinct thematic patterns across Disney and Studio Ghibli films. Overall, Ghibli films exhibited stronger associations with themes such as Grief, Friendship, and Personal Growth, whereas Disney films showed higher thematic relevance for Loneliness, Trauma, and Identity.

Notably, Grief emerged as a more prominent theme in Ghibli, with an average relevance score of 0.8657 compared to 0.6608 in Disney. This supports the idea that Ghibli narratives often explore sorrow and loss with greater emotional depth. In contrast, Loneliness was significantly more central in Disney films (theme score: 0.9161 vs. Ghibli's 0.5750), suggesting it plays a more explicit narrative role, perhaps framed by a journey toward connection or resolution.

Themes such as Romantic Love and Friendship also displayed studio-specific emphases. Romantic Love appeared only in Disney, while Friendship was only detected in Ghibli. Identity and Empathy were more prevalent in Disney films, aligning with character-driven plots focused on self-discovery and moral development. These results highlight how the zero-shot model helped uncover thematic distinctions that sentiment analysis alone could not capture. A numerical comparison of sentiments and themes by studio can be found in appendix F.

7. Conclusion:

This comparative study of the dialogue in animated films by Studio Ghibli and Disney, using computational analysis techniques, has revealed significant differences in dominant themes, gender representation, and narrative structure. Studio Ghibli stands out for addressing themes such as grief, friendship, and personal growth, while Disney focuses more on loneliness, identity, and trauma. These thematic contrasts reflect fundamentally different ways of understanding emotions and human relationships: Ghibli leans toward introspective and community-based narratives, while Disney prioritizes individual resolution and more traditional adventure-driven plots.

One of the most revealing findings was the contrast between Romantic Love and Friendship: romantic love appeared exclusively in Disney films, while friendship was only present in Ghibli's. Rather than being surprising, this reflects broader cultural tendencies, while Western narratives often center romantic resolution as a key emotional goal, Japanese storytelling, as seen in Ghibli, tends to emphasize platonic bonds and collective experience, with romantic feelings expressed more subtly or indirectly. This difference highlights how cultural norms shape emotional expression in animation.

The applied topic modeling showed greater thematic coherence in Ghibli films, particularly around natural and fantastical motifs, reinforcing the studio's well-known environmental aesthetic. Disney, on the other hand, featured language more centered on danger and conflict. Which was totally unexpected for us, and it's possible that the language models were not good enough at distinguishing these differences. In terms of gender representation, a higher frequency of male-associated terms appeared in Disney, while Ghibli displayed a more prominent presence of female protagonists. These linguistic differences suggested not only distinct narrative styles but also divergent positions on gender roles within animated storytelling, which make sense for the type of movies and environments they develop in. We have seen a change in Disney through the years though.

8. Author Contributions:

Everyone: Presentation, Project Updates, Initial Research, Writing report, Research for Topic

Modeling and Sentiment Analysis, Finding Datasets
Francesca: Initial model exploration Sentiment Analysis, Data splitting for Topic Modeling, LDA for Topic Modelling
Violeta: Research for the papers and different possible models to apply, finally use of zero-shot classification for finding more labels for sentiment analysis
Ana: Data preprocessing, research for possible Sentiment Analysis models and final implementation of CardiffNLP's twitter-roberta-base-sentiment model.

9. References:

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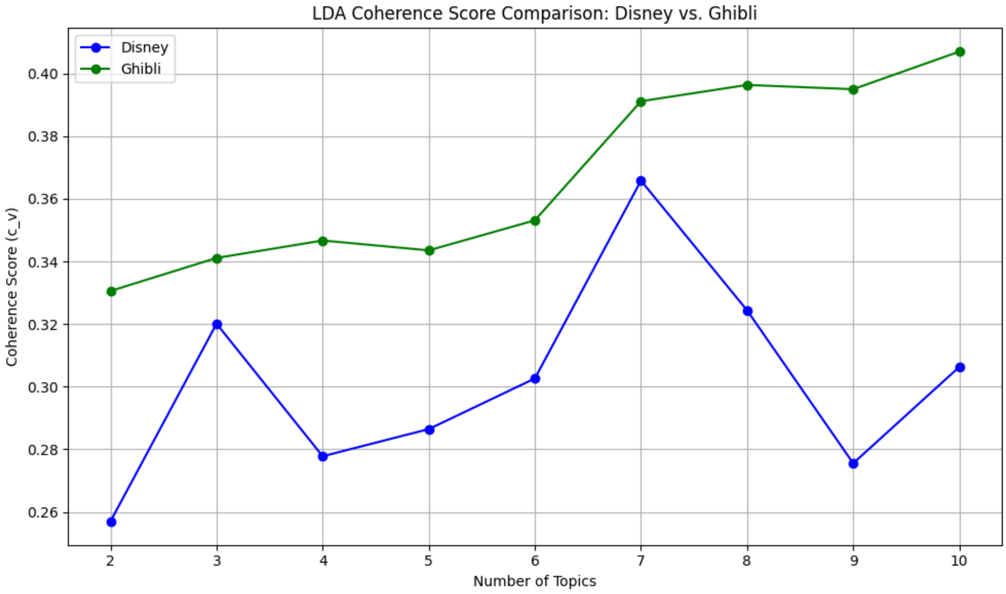
Appendix:

Appendix A:



Appendix B:

Topic Modeling Results: Disney

Topic	Suggested Topic Name	Top Words
0	Royalty & Family	**mother**, year, **baby**, **princess**, life, **mum**, **kingdom**, mine
1	Music & Creativity	**music**, people, sense, part, **piece**, work, **color**, rock
2	Unclear	life, prince, princess, world, friend, star, home, place
3	Heroism & Conflict	**hero, savage**, officer, **predator,** world, kid, god, carrot
4	Games & Competition	**game**, kid, **race**, heart, friend, man, wreck, home
5	Unclear	dream, dig, heart, man, talk, home, work, love
6	Love & Adventure at Sea	heart, love, **island, sea**, people, **boat**, hook, **water**

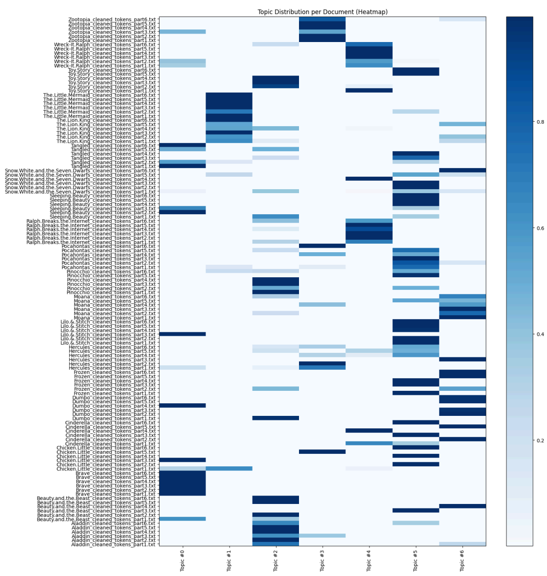
Appendix C:

Topic Modeling Results: Studio Ghibli

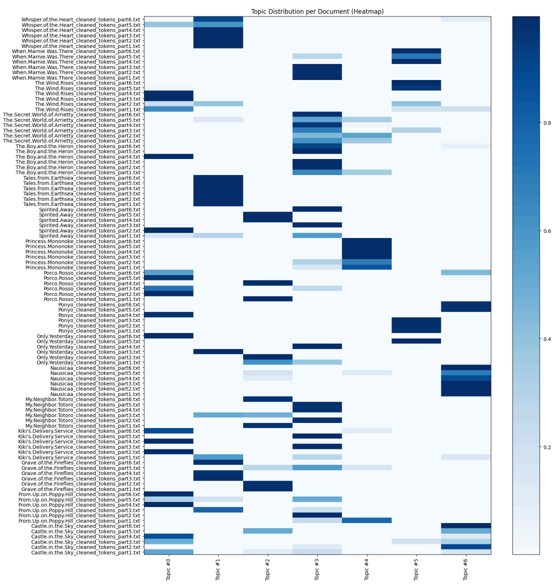
Topic	**Suggested Topic Name**	**Top Words**
0	Work & War	**work**, hand, **plane**, year, **engine, job, war**, witch
1	Everyday Life / Town	**road, home, life, town, country, school**, dad, **work**
2	Family & Childhood	**mama, dad**, tree, school, plane, bath, **house, sister**
3	Home	**home**, mother, house, world, mom, people, work, place
4	Spirit World & Mythology	**spirit**, wolf, **human**, **god**, woman, **life**, men, **head**
5	Unclear	home, design, kurokawa, school, year, night, people, child
6	Nature & Fantasy	princess, **sea,** decay, valley, **fire, wind, stone,** ship

Appendix D:

Disney:



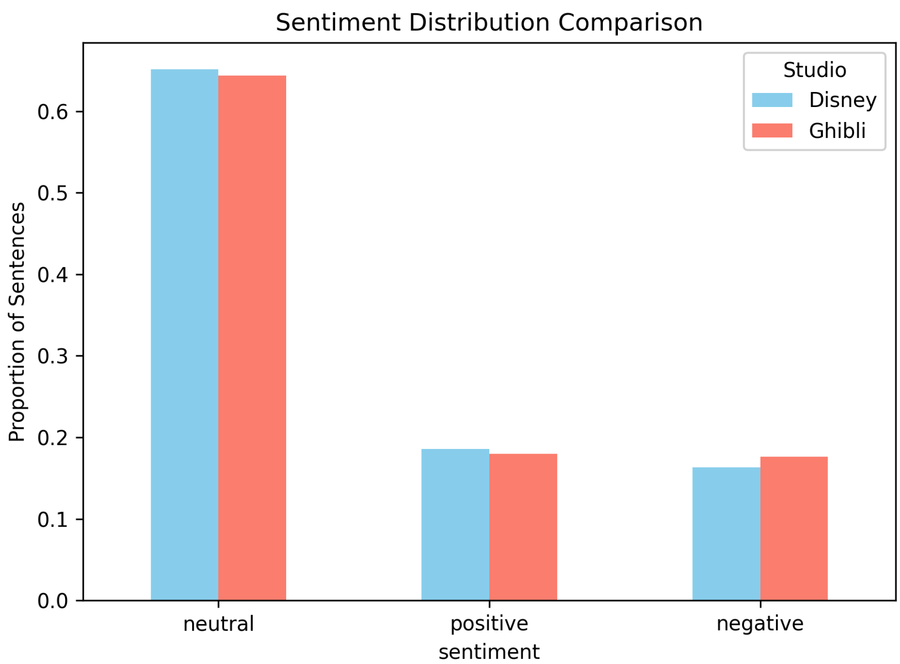
Ghibli:



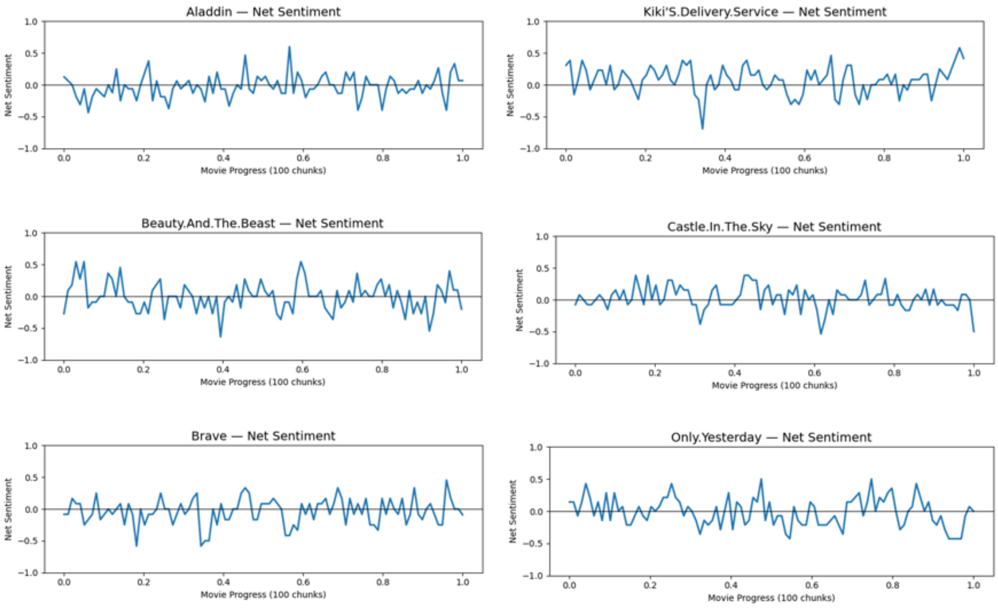
Appendix E:

Word	Disney Probabilities (Topic, Prob.)	Ghibli Probabilities (Topic, Prob.)
man	(0, 0.0055), (1, 0.0084), (2, 0.0026)	(0, 0.0061), (1, 0.0049), (2, 0.0047), (4, 0.0111), (5, 0.0068), (6, 0.0028)
men	(0, 0.0042), (1, 0.0022)	(0, 0.0021), (1, 0.0047), (4, 0.0132)
son	(0, 0.0037), (1, 0.0044), (2, 0.0015)	(0, 0.0036), (3, 0.0024)
prince	(0, 0.0029), (1, 0.0085), (2, 0.0046)	(1, 0.0017), (4, 0.0038)
dad	(0, 0.0019), (1, 0.0035), (2, 0.0015)	(0, 0.0029), (1, 0.0092), (2, 0.0115), (3, 0.0068)
woman	(0, 0.0016), (1, 0.0010)	(0, 0.0038), (1, 0.0044), (2, 0.0044), (3, 0.0024), (4, 0.0150)
daughter	(0, 0.0028), (2, 0.0036)	(3, 0.0025), (4, 0.0025), (5, 0.0022)
princess	(0, 0.0042), (1, 0.0072), (2, 0.0019)	(4, 0.0025), (6, 0.0290)
mom	(2, 0.0031)	(0, 0.0031), (1, 0.0040), (2, 0.0035), (3, 0.0118)

Appendix F:



Appendix G:



Appendix H:

Numerical Comparison of Sentiments and Themes by Studio

Theme/Sentiment	Ghibli (Average Sentiment)	Ghibli (Average Theme)	Disney (Average Sentiment)	Disney (Average Theme)
Resilience	0.4523	0.4976	0.4079	0.3640
Grief	0.5477	0.8657	0.4611	0.6608
Trauma	0.5283	0.4835	0.4517	0.6705
Personal Growth	0.5127	0.5507	0.4285	0.6325
Empathy	0.4964	0.0894	0.4511	0.3291
Romantic Love	N/A	N/A	0.5153	0.3148

Friendship	0.5856	0.5505	N/A	N/A
Family	0.4396	0.7294	0.4454	0.6832
Loneliness	0.3402	0.5750	0.7797	0.9161
Identity	0.4755	0.0567	0.4057	0.4832